

Adding (i), (ii), and (iii), we get: $a + b + c$
 $= K(2a + 2b + 2c) \Rightarrow K = \frac{a+b+c}{2(a+b+c)} = \frac{1}{2}$.

112. $a - 8 = b \Rightarrow a - b = 8$ and $b - a = -8$.

$\therefore |a - b| - |b - a| = |8| - |-8| = 8 - 8 = 0$.

113. $x = \frac{a}{a-1} = 1 + \frac{1}{a-1} = 1 + y$. $\therefore x > y$.

114. a is positive and $a < 1 \Rightarrow \frac{1}{a} > 1$. $\therefore \left(a + \frac{1}{a}\right) > 2$.

115. $\frac{a}{x} + \frac{y}{b} = 1 \Rightarrow \frac{a}{x} = 1 - \frac{y}{b} = \frac{b-y}{b} \Rightarrow \frac{x}{a} = \frac{b}{b-y}$.

$\frac{b}{y} + \frac{z}{c} = 1 \Rightarrow \frac{z}{c} = 1 - \frac{b}{y} = \frac{y-b}{y} \Rightarrow \frac{c}{z} = \frac{y}{y-b} = \frac{-y}{(b-y)}$.

$\therefore \frac{x}{a} + \frac{c}{z} = \frac{b}{(b-y)} - \frac{y}{(b-y)} = \frac{(b-y)}{(b-y)} = 1$.

116. $a^2 + b^2 = 45$... (i)
 and $b^2 + c^2 = 40$... (ii)

Subtracting, we get: $a^2 - c^2 = 5 \Rightarrow (a + c)(a - c) = 5$.

$\therefore (a + c) = 5$ and $(a - c) = 1$.

Solving, we get: $a = 3, c = 2$. Putting $c = 2$ in (ii), we get: $b = 6$.

117. $x + y = 15 \Rightarrow (x + y)^2 = (15)^2 = 225$
 $\Rightarrow x^2 + y^2 + 2xy = 225$
 $\Rightarrow x^2 + y^2 = 225 - 2xy = 225 - 2 \times 56$
 $= 225 - 112 = 113$.

118. $(a - b) = 4 \Rightarrow (a - b)^2 = 4^2 = 16 \Rightarrow a^2 + b^2 - 2ab = 16$
 $\Rightarrow a^2 + b^2 = 16 + 2ab = 16 + 2 \times 2$
 $= 16 + 4 = 20$.

119. $a = bc$ and $c = a - b \Rightarrow a = b(a - b) \Rightarrow a = ab - b^2$
 $\Rightarrow b^2 = ab - a = a(b - 1) \Rightarrow a = \frac{b^2}{b-1}$.

120. $a + b + c = 0 \Rightarrow a = -(b + c) \Rightarrow a^2 = (b + c)^2$.

$\therefore \frac{a^2}{(a^2 - bc)} + \frac{b^2}{(b^2 - ca)} + \frac{c^2}{(c^2 - ab)}$
 $= \frac{(b+c)^2}{(b+c)^2 - bc} + \frac{b^2}{b^2 + c(b+c)} + \frac{c^2}{c^2 + b(b+c)}$
 $= \frac{(b+c)^2}{b^2 + c^2 + bc} + \frac{b^2}{b^2 + c^2 + bc} + \frac{c^2}{b^2 + c^2 + bc}$
 $= \frac{b^2 + c^2 + 2bc + b^2 + c^2}{b^2 + c^2 + bc} = \frac{2(b^2 + c^2 + bc)}{b^2 + c^2 + bc} = 2$.

121. $\frac{x^2 - y^2}{x^2 + y^2} = \frac{\left(\frac{x^2}{y^2}\right) - 1}{\left(\frac{x^2}{y^2}\right) + 1} = \frac{\left(\frac{x}{y}\right)^2 - 1}{\left(\frac{x}{y}\right)^2 + 1} = \frac{\left(\frac{a+2}{a-2}\right)^2 - 1}{\left(\frac{a+2}{a-2}\right)^2 + 1}$
 $= \frac{(a+2)^2 - (a-2)^2}{(a+2)^2 + (a-2)^2} = \frac{4 \times a \times 2}{2(a^2 + 4)} = \frac{4a}{a^2 + 4}$.

122. $3x + 7 = 7x + 5 \Rightarrow 7x - 3x = 2 \Rightarrow 4x = 2 \Rightarrow x = \frac{1}{2}$.

Now, $3x + 7 = x^2 + P \Rightarrow \frac{3}{2} + 7 = \frac{1}{4} + P$
 $\Rightarrow P = \frac{17}{2} - \frac{1}{4} = \frac{33}{4} = 8\frac{1}{4}$.

123. $\frac{2a+b}{a+4b} = 3 \Rightarrow 2a+b = 3(a+4b) \Rightarrow a = -11b$.

$\frac{a+b}{a+2b} = \frac{-11b+b}{-11b+2b} = \frac{-10b}{-9b} = \frac{10}{9}$.

124. $(2a + 3b)(2c - 3d) = (2a - 3b)(2c + 3d)$
 $\Rightarrow \frac{(2a + 3b)}{(2a - 3b)} = \frac{(2c + 3d)}{(2c - 3d)}$

$\Rightarrow \frac{2\left(\frac{a}{b}\right) + 3}{2\left(\frac{a}{b}\right) - 3} = \frac{2\left(\frac{c}{d}\right) + 3}{2\left(\frac{c}{d}\right) - 3} \Rightarrow \frac{a}{b} = \frac{c}{d}$.

125. $(a + b + 2c + 3d)(a - b - 2c + 3d)$
 $= (a - b + 2c - 3d)(a + b - 2c + 3d)$
 $\Rightarrow [(a + b) + (2c + 3d)][(a - b) - (2c - 3d)]$
 $= [(a - b) + (2c - 3d)][(a + b) - (2c + 3d)]$
 $\Rightarrow (a + b)(a - b) - (a + b)(2c - 3d) + (a - b)(2c + 3d)$
 $- (2c + 3d)(2c - 3d)$
 $= (a - b)(a + b) - (a - b)(2c + 3d) + (a + b)(2c - 3d)$
 $- (2c + 3d)(2c - 3d)$
 $\Rightarrow (a + b)(2c - 3d) = (a - b)(2c + 3d)$
 $\Rightarrow 2ac - 3ad + 2bc - 3bd = 2ac + 3ad - 2bc - 3bd$
 $\Rightarrow 4bc = 6ad \Rightarrow 2bc = 3ad$.

126. $\frac{x+y}{x-y} + \frac{(x+y)^2}{x^2 - y^2} = \frac{x+y}{x-y} \times \frac{(x^2 - y^2)}{(x+y)^2}$
 $= \frac{(x^2 - y^2)}{(x-y)(x+y)} = \frac{(x^2 - y^2)}{(x^2 - y^2)} = 1$.

127. $\frac{a^2 - b^2 - 2bc - c^2}{a^2 + b^2 + 2ab - c^2} = \frac{a^2 - (b^2 + 2bc + c^2)}{(a^2 + b^2 + 2ab) - c^2} = \frac{a^2 - (b+c)^2}{(a+b)^2 - c^2}$
 $= \frac{(a+b+c)(a-b-c)}{(a+b+c)(a+b-c)} = \frac{a-b-c}{a+b-c}$.

128. $x + y + z = 0 \Rightarrow y = -(x + z)$
 $\therefore x^2 + xy + y^2 = x^2 + x(-x - z) + (x + z)^2$
 $= x^2 - x^2 - xz + x^2 + z^2 + 2xz$
 $= x^2 + z^2 + zx$.

129. $(s - a)^2 + (s - b)^2 + (s - c)^2 + s^2$
 $= (s^2 + a^2 - 2sa) + (s^2 + b^2 - 2sb) + (s^2 + c^2 - 2sc) + s^2$
 $= 4s^2 + (a^2 + b^2 + c^2) - 2s(a + b + c)$
 $= (2s)^2 + (a^2 + b^2 + c^2) - (a + b + c)(a + b + c)$
 $= (a + b + c)^2 + (a^2 + b^2 + c^2) - (a + b + c)^2$
 $= a^2 + b^2 + c^2$.

130. $[(x + y)^2 - z^2] + [(y + z)^2 - x^2] + [(z + x)^2 - y^2]$
 $= 4 + 9 + 36$
 $\Rightarrow (x + y + z)(x + y - z) + (x + y + z)$
 $(y + z - x) + (x + y + z)(z + x - y) = 49$
 $\Rightarrow (x + y + z)[(x + y - z) + (y + z - x) + (z + x - y)] = 49$
 $\Rightarrow (x + y + z)(x + y + z) = 49 \Rightarrow (x + y + z)^2 = 49$
 $\Rightarrow (x + y + z) = \pm 7$.